

Installation Manual

Power Quality Improvement and Harmonic Mitigation



21st Century Power Quality Improvement





SAFETY

WARNING!

**THE EQUIPMENT COVERED BY THIS PUBLICATION MUST BE INSTALLED
FOR A SPECIFIC APPLICATION .**

**ONLY QUALIFIED ELECTRICIANS LICENSED AND DULY AUTHORIZED
WHO ARE CAPABLE OF FOLLOWING SPECIFIC INSTRUCTIONS AND CODES OF COMPLIANCE
SHOULD HANDLE THE INSTALLATION.**

THE EQUIPMENT MUST BE INSTALLED, OPERATED AND MAINTAINED BY QUALIFIED PERSONNEL .

**IT IS MANDATORY THAT INSTALLATION PERSONNEL ARE CERTIFIED
AND ARE THOROUGHLY TRAINED ELECTRICIANS WHO UNDERSTAND POTENTIAL HAZARDS
INVOLVED WITH ELECTRICAL INSTALLATIONS .**

MUST BE INSTALLED IN COMPLIANCE WITH NEC AND LOCAL ELECTRICAL CODES

**IF ELECTRICAL CODES ARE NOT COMPLIED THE PRODUCT WARRANTY
WILL BE NULL AND VOID**

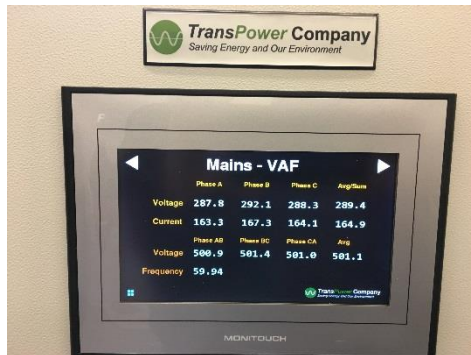
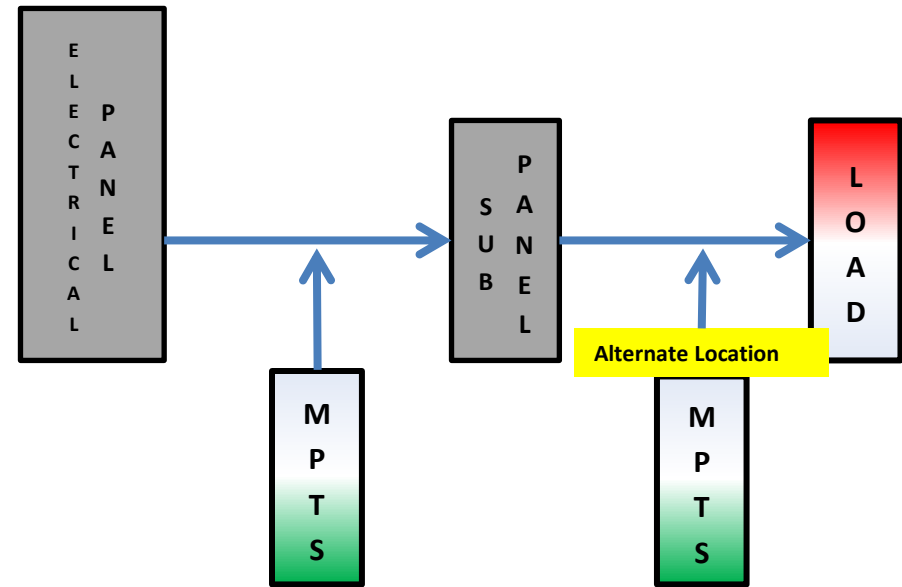
MPTS

21st Century Power Quality Improvement and Harmonics Mitigation

Maximum Power Transfer Solution (with Harmonic Filters suitable for 3 wire delta or 4 wire Y connections MPTS) enhances electrical energy efficiency, lowers total electrical energy consumption, compliments environmental sustainability and reduces cost.

The solution is aimed towards efficient utilization of electric power for loads. Power Factor correction is simultaneously achieved.

This solution almost eliminates wasted electrical power and subsequently reduces total energy consumption by significant levels depending on the electrical system's inefficiency.



MPTS - Installation Manual

This manual is intended as a general guide for the installation, commissioning, maintenance, and troubleshooting of MPTS units.

This manual is written for qualified and licensed personnel exclusively.

It is not a substitute for professional training, safety and hazard procedures, or for handling industrial equipment.

After going through the contents of this manual, if there are any doubts with regard to connectivity or any other subject, they should be clarified with the supplier.

This is essential to avoid electric shock or fire resulting in personal injury and damage to equipment. Improper connection, handling, or installation may cause malfunction and potential damage.

Devices must be installed in compliance with NEC and Local electrical codes.

**WHEN ELECTRICAL CODES ARE NOT COMPLIED WITH FOR INSTALLATION
THE PRODUCT WARRANTY WILL BE NULL AND VOID**

General Overview

Majority of the loads in commercial as well as industrial applications have very rapidly changing loads due to the nature of their operation.

The overall nature of the electrical loads is complex inductive load even though there is some resistive and capacitive components in all equipment. This complex inductive load causes a lagging power factor in electrical systems and network resulting in high electrical demand.

Inefficient electrical systems and equipment draw excessive current and power (KVA) than what is required to perform useful work.

Utilities have to design their distribution network to provide for the electrical demand and make provisions in the capacity for inefficient electrical systems and equipment.

Use of MPTS devices on commercial and industrial applications decreases the inefficiencies resulting in reduced total electrical demand and current consumption.

In addition, use of MPTS releases electrical capacity in installed switchgear, decrease voltage fluctuations due to high KVA draw, lowers heating of switchgear and transformers in the customer's electrical network.

Check list for Receiving the MPTS shipment

Check list for receiving Shipments

Do not sign delivery paperwork (BOL) until you have checked of the following

		Circle One		
1	Did you check the cargo before unloading?	Yes	or	No
2	Did you notice any damage	Yes	or	No
3	Did you supervise the unloading from the truck	Yes	or	No
4	Did you walk around the crate to look for any damage	Yes	or	No
5	Did you look on top of the Crate	Yes	or	No
6	Did you notice any damage to the crate	Yes	or	No
7	If any damage seen, did you write it down on the delivery paperwork?	Yes	or	No
8	Did you point out the damage to the Driver of the truck?	Yes	or	No
9	Did you take photographs of the Damage <i>Please make sure you take photographs - this is very important to file a claim.</i>	Yes	or	No
10	Did you write down the damage on the Delivery paperwork above your signature <i>Note down the following above your signature on the Delievery paperwork.</i> <i>Received shipment with external Damage, will confirm internal damage after un Packaging</i>	Yes	or	No
<i>If there was any damage- Please arrange to open the crate immediately to evaluate any damage to the equipment inside.</i>				
<i>This is important because if we have to report any damage this has to be done within 24 hours.</i>				
Make sure you have the Driver acknowledge the Damage on paperwork - sign off beside your comment .& keep a copy.				
Date:		Time:		
Cargo Receiving Person:		Driver:		
Name				
Phone number				
Email				



Damage in shipping

Unpacking and Inspection

MPTS devices are always transported in the upright position on crated pallets.

The MPTS device is fixed to a wooden pallet with bolts for shipping. It is necessary to use a fork lift in order to move the unit; the unit should be lifted from the base. Caution should be exercised not to damage the unit while using material handling equipment

When Equipment is delivered to the customer site, inspect the shipment for any external damage to the shipment/ exterior crate.

Even if you see a slight damage, mention “slight external damage, possible internal damage” above your signature in the receiving / delivery document

In case of any damage, please take photos of the damaged area as these are necessary for insurance claims. Immediately inform your freight company to lodge a claim with the carrier.

Carefully open the wooden Crate. Re-inspect to make sure that there is no damage to the unit. In case there is any damage, please take photos of that area.

The MPTS units are packed with a strong recyclable wooden crate for safe transportation.



Damage in shipping

Storing the MPTS

MPTS devices are always transported in the upright position on crated pallets. The MPTS device is fixed to a wooden pallet with bolts for shipping. It is necessary to use a fork lift in order to move the unit; the unit should be lifted from the base. Caution should be exercised not to damage the unit while using material handling equipment.

Store the MPTS in a secure and conditioned space.
Treat the device like a computer.



Unloading, Unpackaging, and putting Castors on MPTS Units

It is necessary to use a forklift in order to unload the MPTS from the truck. The unit should be lifted from the base. Caution should be exercised not to damage the unit while using material handling equipment. The MPTS device is fixed to a wooden pallet with bolts for shipping.

When Equipment is delivered to the customer site, Even if you see a slight damage to the crate, mention “slight external damage, possible internal damage” above your signature in the receiving / delivery document / BOL by the freight company.

In case of any damage, please take photos of the damaged area as these are necessary for insurance claims. Immediately inform your freight company to lodge a claim with the carrier.

Carefully open the wooden Crate. Use the right tools to unbolt the crate packaging.

Re-inspect to make sure that there is no damage to the MPTS. In case there is any damage, please take photos of that area. These have to be submitted as soon as possible.

Please store the crate panels and bolts if you have a need to re ship/re package/ or move the MPTS by truck.

Unloading, Unpackaging, and putting on Castors on MPTS



Unloading



Un bolt and
crate panels removed



Putting Castors on the MPTS



In position for connections

MPTS Dimensions and Weight			
Item	Description	Dimensions	Weight
	Models	Inches	Pounds
1	MPTS H240A / 480V / 60Hz	32 x 43 x 56	950 lbs.
2	MPTS H490A / 480V / 60Hz	36 x 63 x 63	1870 lbs.
3	MPTS H720A / 480V / 60Hz	36 x 63 x 79	2800 lbs.
<i>The information presented in this document or publication has been checked for reliability; however no responsibility is assumed for inaccuracies. TransPower Company, Inc strives for continuous improvement in published documents, correcting inaccuracies as they become known to us. The reader should not rely upon this document or publication for any legal purpose whatsoever. If you believe this document or publication to contain inaccuracies please report it directly to the company.</i>			

Specifications	MPTS H240	MPTS H490	MPTS H720
One MPTS for:	up to 600A Service	up to 1200A Service	up to 1200A Service
Multiple MPTS units	can be installed in the electrical network/panels/sub panels based on requirement		
Current Rating:	240 Amperes	490 Amperes	720 Amperes
KVAr Rating:	200	400	600
Voltage Input:	480 VAC 60Hz		
Electrical Supply:	Three Phase		
Configuration : 3Phase-3 wire system (a,b,c & g) or 4 Wire system (a,b,c,n,& g):	3 Phase - 3 or 4 Wire System		
Cable Entry:	Top Entry		
Control Operation:	Automatic switching based on load requirements.		
Controller:	Electronic control system using a Microcontroller		
HMI Displayed Data:	Voltage, Current, KVA, KVAh, KW, KWh, Hz, KVAh, PF		
HMI Display Meter:	Indicates RMS values		
Main Protection:	Short circuit and Over- load protection		
Bank Protection:	Short circuit and Over- load protection		
Construction :	Robust Standalone metal enclosure (indoor application only)		
Dimensions : (WidthxDepthxHeight):	28 x 40 x 56 Inches	34 x 63 x 63 Inches	34 x 63 x 79 Inches
Estimated Weight:	950 lbs.net, 1300 packaged	1870 lbs.net, 2400 lbs. packaged	2800 lbs.net, 3400 lbs. packaged
Mounting:	Floor Mounting with Foundation Bolts.		
Enclosure:	Powder coated corrosion resistant sheet steel enclosure (RAL 7035)		
Transient protection:	Provided		
Power Monitoring:	Yes- built in	Yes- built in	Yes- built in
Remote Data Logging	Yes- built in	Yes- built in	Yes- built in
Power Analyzer and Charts:	Available with Remote Data Logging		
Real time Monitoring:	Graphs and Data values available with and without MPTS		
Programming:	Self programing - no programming needed - connect and turn ON		
Fail Safe:	Operates in parallel - no interruption of power to the load when the MPTS is active or inactive		
Specifications:	may be updated with improvements		
Suggested Cable Rating:	Per regulatory requirement		
Main incoming Breaker Rating:	400 Amps	800 amps	1000 Amps
Suggested Cable Rating:	As per NEC		

240A/480V/3Ph/200KVA



Dimensions & Weight

Front = 32 inches

Dept = 40 Inches

Height = 56 inches

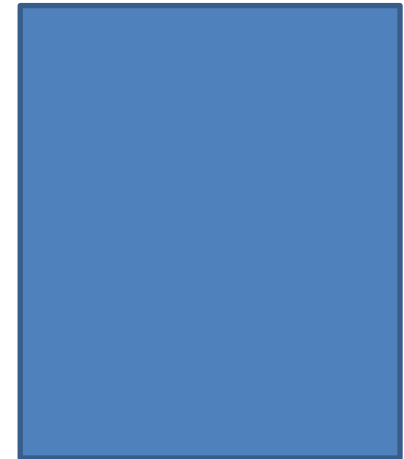
Weight = 950 lbs.

32 Inches



Front

43 Inches



Side

56 Inches
height

240A/480V/3Ph/200KVAR**Specifications**

Model	: MPTS-H 240 - external CTs
Rated Demand Reduction:	50 Kilowatts
KVAr Rating	: 200
Voltage Input	: 480 VAC 60Hz
Supply	: Three Phase
Configuration	: 3 Phase - 3 or 4 Wire
Cable Entry	: Top Entry
Control Operation	: Automatic switching based on load requirements.
Controller	: Electronic control system using a Microcontroller
Displayed Data	: Voltage, Current, KVA, KVAh, KW, KWh, Hz, KVAr, PF
Display Meter	: Indicates RMS values.
Main Protection	: Short circuit and Over- load protection
Bank Protection	: Short circuit and Over- load protection
Construction	: Robust Standalone metal enclosure (indoor application only)
Dimensions	: 32 x 43 x 56 Inches (Width x Depth x Height)
Estimated Weight	: 950 lbs. net, 1300 packaged
Mounting	: Floor Mounting with Foundation Bolts.
Enclosure	: Powder coated corrosion resistant sheet steel enclosure (RAL 7035)
Transient protection	: Provided
Specifications	: may be updated with improvements



Suggested Cable Rating Per regulatory requirement	Main incoming Breaker Rating 400 Amps	Suggested Cable Rating As per NEC
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490A/480V/3Ph/400KVAR**Dimensions & Weight**

Front = 36 inches

Dept = 63 Inches

Height = 63 inches

Weight = 1870 lbs.

36 Inches



Front

63 Inches



Side

63 Inches

490A/480V/3Ph/400KVAR**Specifications**

Model	: MPTS-H 490 - external CTs
Rated Demand Reduction:	100 Kilowatts
KVAR Rating	: 400
Voltage Input	: 480 VAC 60Hz
Supply	: Three Phase
Configuration	: 3 Phase - 3 or 4 Wire
Cable Entry	: Top Entry
Control Operation	: Automatic switching based on load requirements.
Controller	: Electronic control system using a Microcontroller
Displayed Data	: Voltage, Current, KVA, KVAh, KW, KWh, Hz, KVAR, PF
Display Meter	: Indicates RMS values.
Main Protection	: Short circuit and Over- load protection
Bank Protection	: Short circuit and Over- load protection
Construction	: Robust Standalone metal enclosure (indoor application only)
Dimensions	: 36 x 63 x 63 Inches (Width x Depth x Height)
Estimated Weight	: 1870 lbs. net, 2400 lbs. packaged (=± 3%)
Mounting	: Floor Mounting with Foundation Bolts.
Enclosure	: Powder coated corrosion resistant sheet steel enclosure (RAL 7035)
Transient protection	: Provided
Specifications	: may be updated with improvements



Suggested Cable Rating Per regulatory requirement	Main incoming Breaker Rating 800 Amps	Suggested Cable Rating As per NEC
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720A/480V/3Ph/600KVA_r

Dimensions & Weight

Front = 36 inches

Dept = 63 Inches

Height = 79 inches

Weight = 2800 lbs.



36 Inches



Front

63 Inches



Side

79 Inches

720A/480V/3Ph/600KVA_r**Specifications**

Model	: MPTS-H 720 - external CTs
Rated Demand Reduction:	150 Kilowatts
KVA _r Rating	: 600
Voltage Input	: 480 VAC 60Hz
Supply	: Three Phase
Configuration	: 3 Phase - 3 or 4 Wire
Cable Entry	: Top Entry
Control Operation	: Automatic switching based on load requirements.
Controller	: Electronic control system using a Microcontroller
Displayed Data	: Voltage, Current, KVA, KVAh, KW, KWh, Hz, KVA _r , PF
Display Meter	: Indicates RMS values.
Main Protection	: Short circuit and Over- load protection
Bank Protection	: Short circuit and Over- load protection
Construction	: Robust Standalone metal enclosure (indoor application only)
Dimensions	: 36 x 63 x 79 Inches (Width x Depth x Height)
Estimated Weight	: 2800 lbs. net, 3400 lbs. packaged (=± 3%)
Mounting	: Floor Mounting with Foundation Bolts.
Enclosure	: Powder coated corrosion resistant sheet steel enclosure (RAL 7035)
Transient protection	: Provided
Specifications	: may be updated with improvements

Suggested Cable Rating Per regulatory requirement	Main incoming Breaker Rating 1000 Amps	Suggested Cable Rating As per NEC
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Special Precautions

The following are mandatory safety measures when installing or using the equipment

Measure the electrical parameters to ensure that the unit selected is appropriately sized.

The units are designed for indoor use only and should be installed in a clean, secure, and dry environment.

During transportation from the manufacturing plant to the customer's premises, there is always a chance that some connections within the unit may become loose. It is mandatory to internally inspect the unit and tighten any loose connections prior to installation and start-up.

It is recommended to open the side panels, visually inspect and check internal condition.

Depending on the environment and site conditions, the unit should be periodically inspected internally by a qualified electrician for normal routine maintenance.

Pre-Installation Assessment

During transportation from manufacturing to customer's premises, there is always a chance of some connections within the MPTS unit becoming loose. Open the panels to inspect inside the unit.

It is mandatory to internally inspect the unit and tighten any loose connections. Do not overtighten and termination or connection.

In case there is insufficient light to conduct a good internal inspection, use a strong torch light or similar light to closely inspect wiring and other connections for looseness.

The following are some typical inspections to be conducted:

	Location	Description
1.	Terminal Connectors	Check for tightness of all connections
2.	Current Transformers	Check for tightness of output connections
3.	Capacitor terminals	Check for tightness of Power wiring connections- it is important not to over tighten as this may result in damage and failure
4.	Reactor terminals	Check the tightness of the power wiring connections, also check the thermostat terminal tightness this is important that any loose contact of these terminals cause trip in the unit.
5.	PCB Boards	Ensure all control wires and connectors or ribbon cables are tightly fitted
6.	General Visual Check	Visually browse through the system internals to ensure that there are no loose wires or unsecured components

MPTS - Devices

Step by Step Connections and installation

Place the MPTS unit rigidly in a clean, secure, and dry suitable environment as close to the distribution panel as possible with sufficient space to connect the terminal section.

Open the MPTS panel doors and visually inspect to ensure that nothing has become loose during transportation and handling. If there is any evidence of damage or loose wires, repair or replace prior to proceeding further.

Select Side or Top cable entry gland plate based on site requirement and disassemble the gland plate by removing fasteners

Drill appropriate size hole for Cable entry and use suitable sized rubber Grommets to ensure adequate safety of cable insulation

All cables to the MPTS should be routed through the rubber grommets.

Turn “OFF” the power to the distribution panel from where the MPTS will be connected.

The main breaker of distribution panel and breakers inside the MPTS should be in ‘OFF’ position.

Step by Step Connection - continued

External CTs

Select the CT size based on the capacity of the power supply breaker

The CT's have to be mounted on the main incomer bus bar/cable on L1, L2, and L3 phases with the correct direction of the CT (**check the current flow direction marked (P1 and P2) on the respective CT, P1 is the current entry**) as shown in the schematic. This is critical to effectively operate the MPTS

CT position should be in such a way that both load current and current to MPTS are flowing through the CTs.

The CT terminals (**usually marked S1 and S2**) should be connected to the CT terminal block within the MPTS unit in exactly the same sequence.

Check the schematic for further clarity.

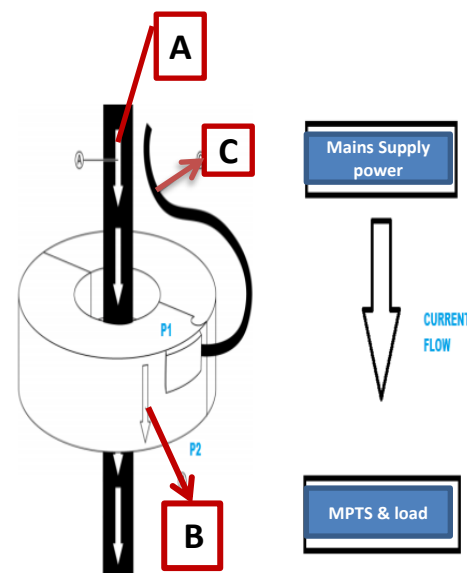
Example: The CT on Line one(1) will be connected to the Mains CT Input Termination inside the MPTS to S1 and S2

Similarly connect incoming supply main cables in the right sequence. Use approved cable with suitable sizing for connection from the distribution panel to the MPTS MCCB as shown in the connection schematic.

Connect proper Ground for MPTS with approved cable appropriately sized as per code.

When external CT's are connected note down the CT ratio. This information may be required for input into the controller module/display meter.

Direction of current flow
in the cable should match **B** On the CT



Legend:

A. Power source cable and power flow

B. CT Mounting orientation indicator arrow

C. CT cables to MPTS

Step by Step Connection - continued

External CTs

Select the CT size based on the capacity of the power supply breaker

Split Core CTs on the Mains Bus Bar or Lines a b and c

External CTs - specifications

Primary Current rating of the
Mains Breaker or 1.5 x Maximum
measured current on the bus bar or cable

Rated Primary Current
400A, 600A, 800A, 1000A

Rated Secondary Current 5A

Burden 2.5VA

Accuracy 0.5%

Withstand Voltage 3,000Vac

Frequency Range 50-400Hz

Operating Temperature -15°C - 60°C

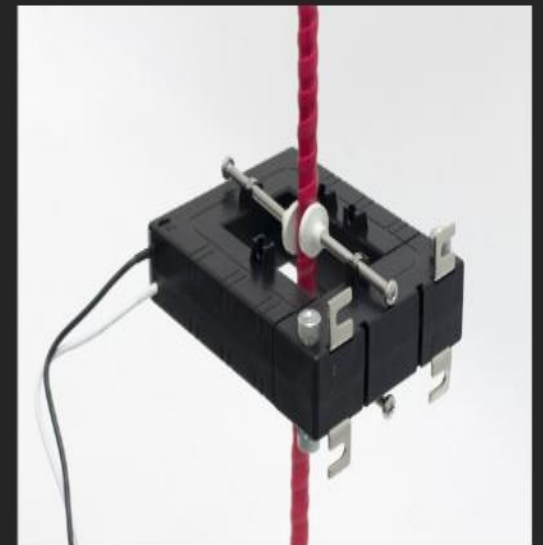
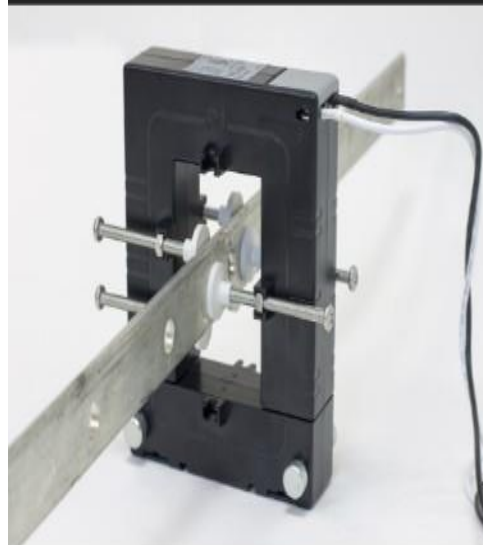
Standard IEC 60044-1

Case Material

UL flame retardant rating 94-V0

Certifications UR, CE approved
& RoHS compliant

Securely mount on cable or bus



Step by Step Connection - continued

External CTs

Select the CT size based on the capacity of the power supply breaker

External CTs - specifications
Primary Current rating of the
Mains Breaker or 1.5 x Maximum
measured current on the bus bar or cable

Rated Primary Current
400A, 600A, 800A, 1000A

Rated Secondary Current 5A

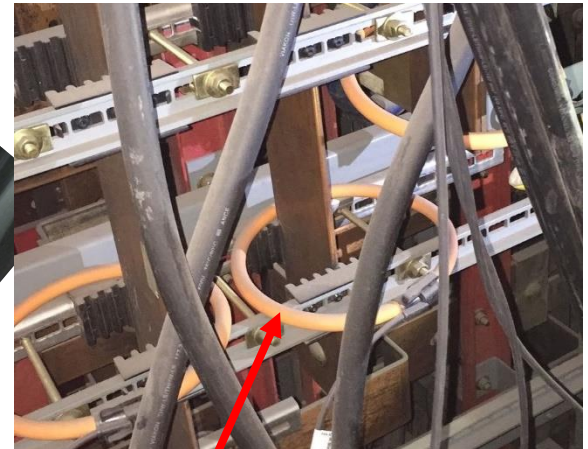
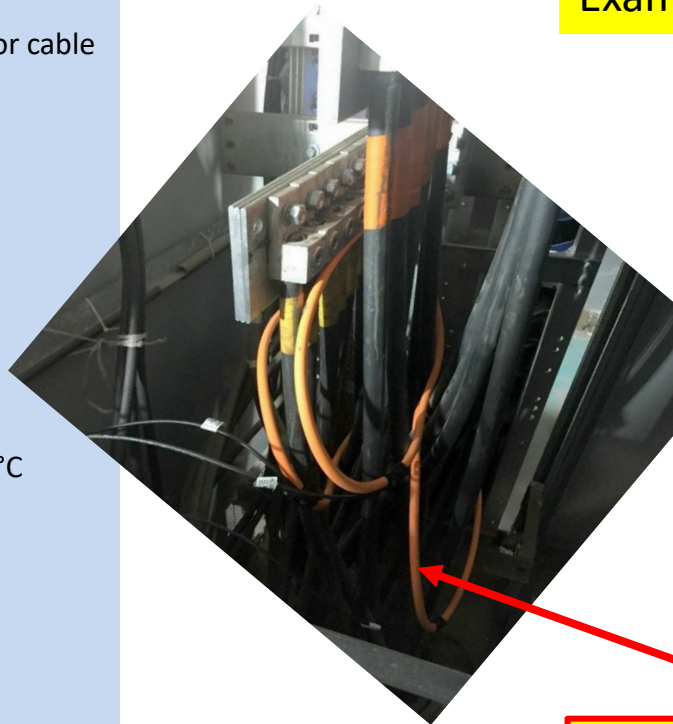
Burden 2.5VA
Accuracy 0.5%
Withstand Voltage 3,000Vac
Frequency Range 50-400Hz
Operating Temperature -15°C - 60°C
Standard IEC 60044-1

Case Material
UL flame retardant rating 94-V0

Certifications UR, CE approved
& RoHS compliant

Rope CTs or Rogowski Coils on the Mains Bus Bar

Examples



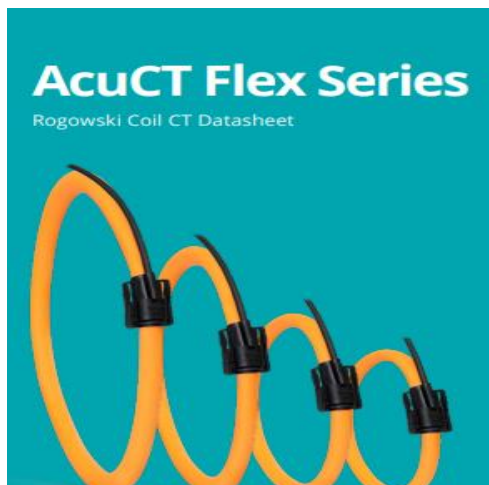
Rogowski Coils / Rope CTs

Step by Step Connection - continued

External CTs

Select the CT size based on the capacity of the power supply breaker

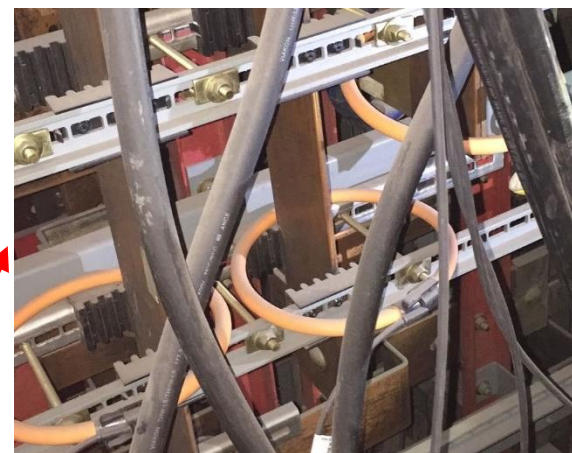
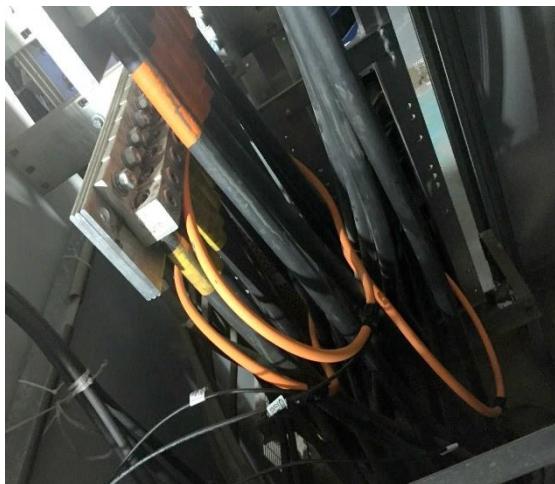
Rope CTs or Rogowski Coils on the Mains Bus Bar



ACCUEVERGY

Specifications

RATED CURRENT	
Current Range	5A-50,000A
Accuracy	0.5% - IEEE C57.13 class 0.6
MECHANICAL/ENVIRONMENTAL	
Form Factor	Flexible Rope Like CT
Window Size	RCT16 - 106.0mm (4.17")
	RCT24 - 178.0mm (7.01")
	RCT36 - 271.0mm (10.67")
	RCT47 - 369.0mm (14.53")
External Diameter	RCT16 - 143.0mm (5.63")
	RCT24 - 207.0mm (8.13")
	RCT36 - 302.0mm (11.89")
	RCT47 - 398.0mm (15.66")
Length of Coil	RCT16 - 400.0mm (15.75")
	RCT24 - 600.0mm (23.62")
	RCT36 - 900.0mm (35.43")
	RCT47 - 1200.0mm (47.24")



Examples

Rogowski Coils / Rope CTs

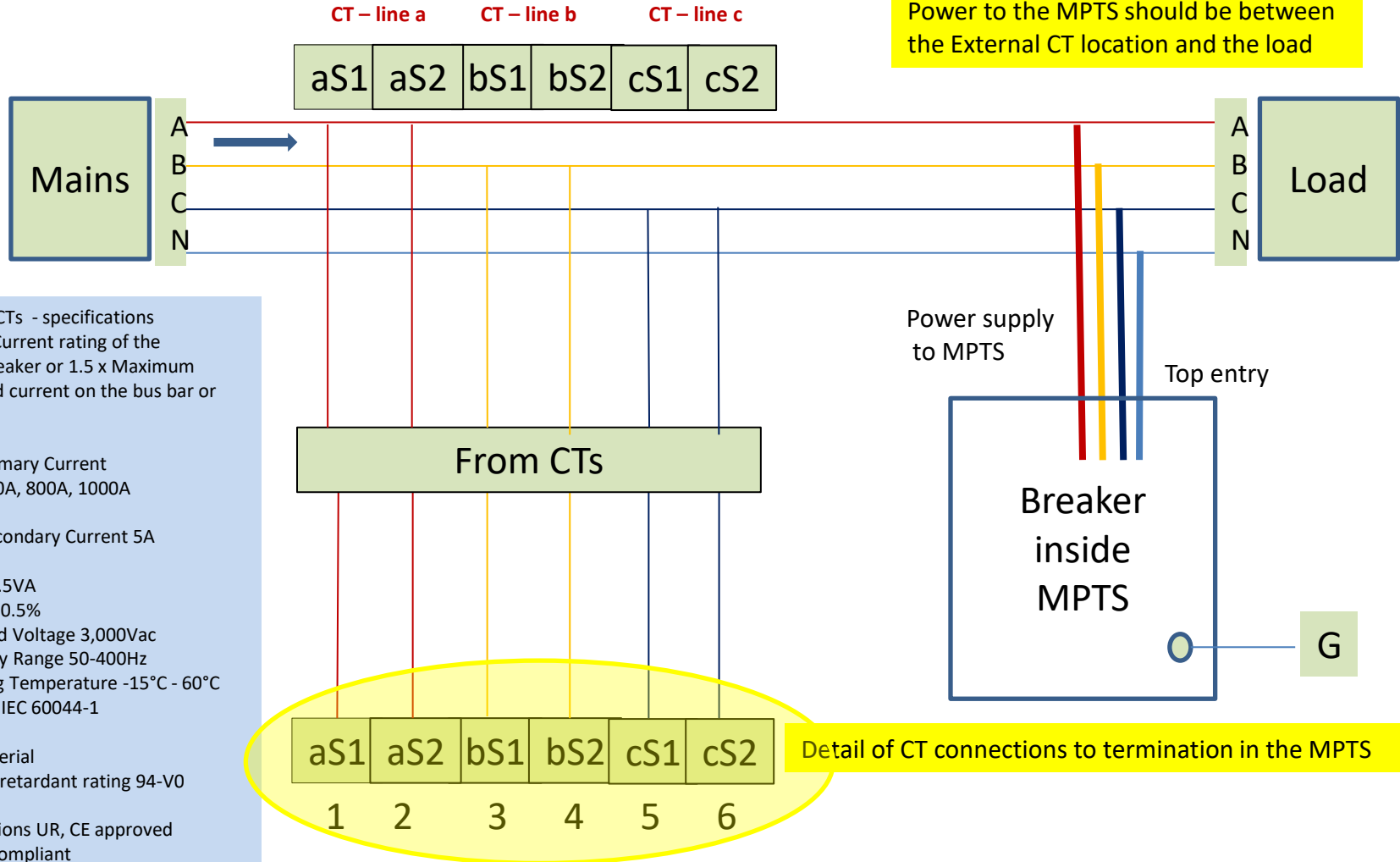
MPTS - Devices

External CTs are chosen based on site conditions
The CT Ratio should be entered in the meter on the MPTS to match the actual value of current in the main cable to the current on the MPTS display

MPTS Wiring details

MPTS location /Power connection

Power to the MPTS should be between the External CT location and the load

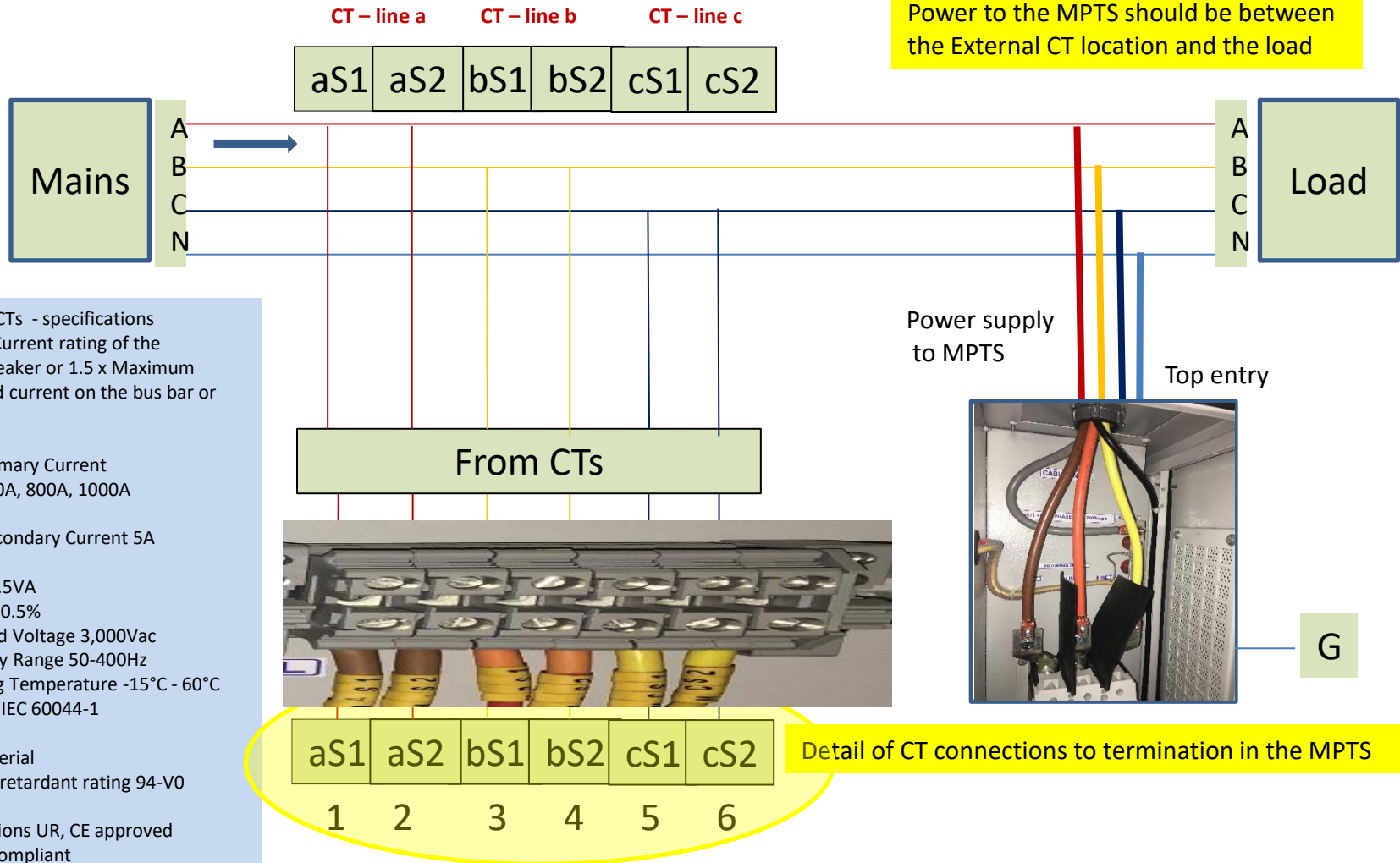


Wire Sizes should comply with NEC and local codes – the minimum current carrying capacity should match the rated current and breaker rating of the MPTS

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The CT Ratio should be entered in the meter on the MPTS to match the actual value of current in the main cable to the current on the MPTS display

MPTS - Devices

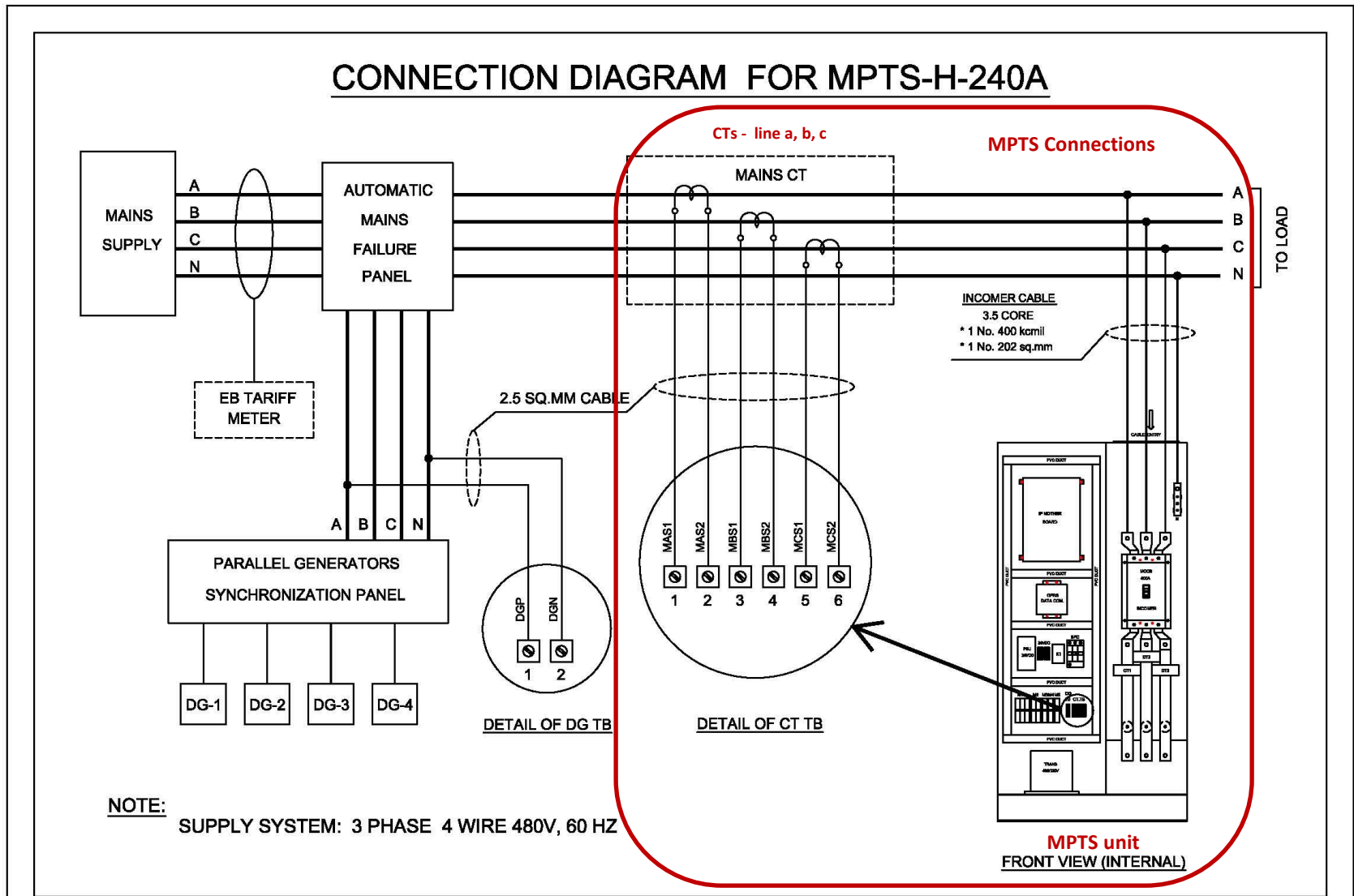
MPTS Wiring details



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MPTS - Devices

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The CT Ratio should be entered in the meter on the MPTS

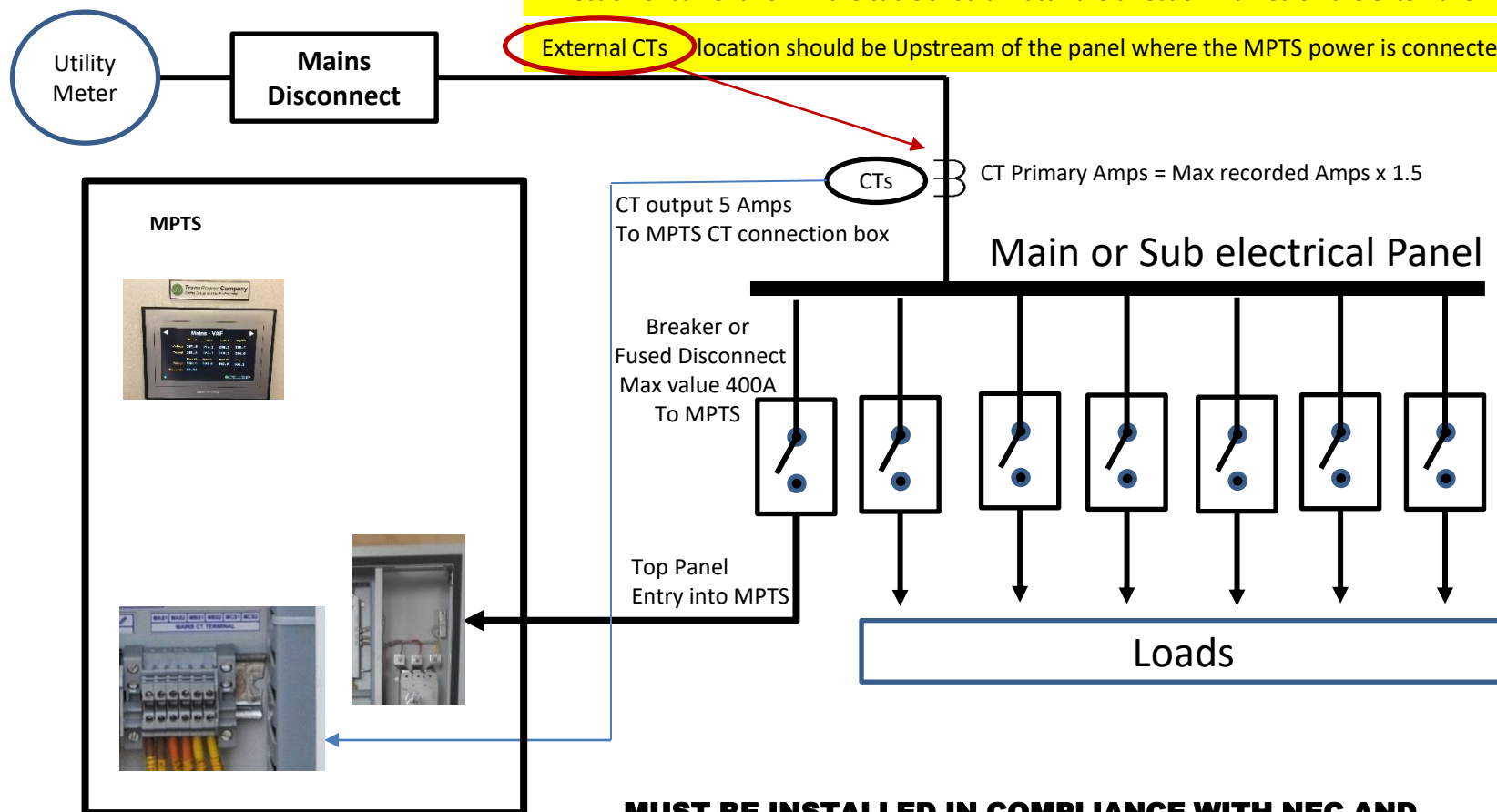


Wire Sizes should comply with NEC and local codes – the minimum current carrying capacity should match the rated current and breaker rating of the MPTS

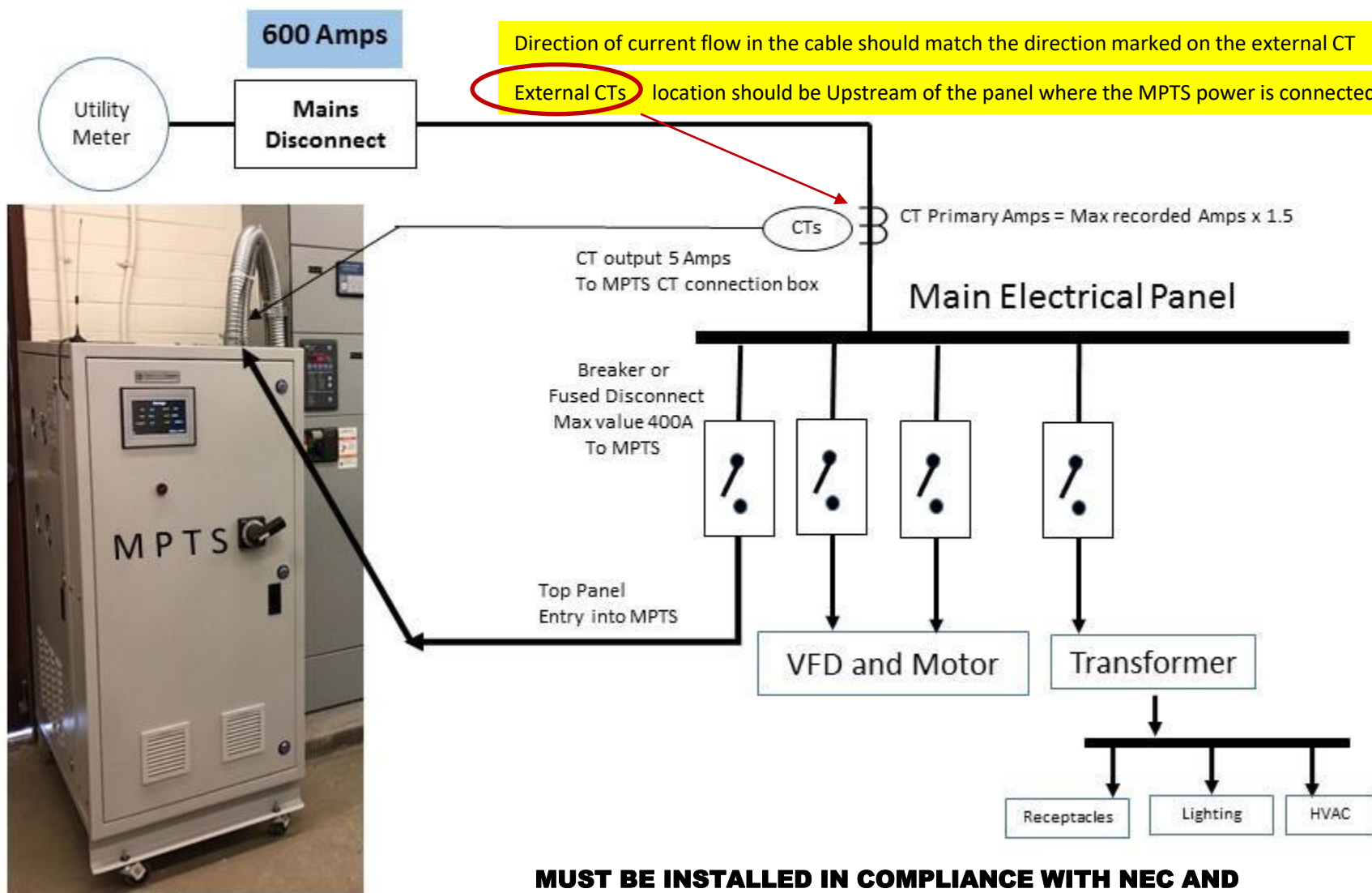
240A/480V/3Ph/200KVA_r

Direction of current flow in the cable should match the direction marked on the external CT

External CTs location should be Upstream of the panel where the MPTS power is connected

**MUST BE INSTALLED IN COMPLIANCE WITH NEC AND
LOCAL ELECTRICAL CODES'**

240A/480V/3Ph/200KVA_r – Power Quality Improvement and Harmonic Mitigation



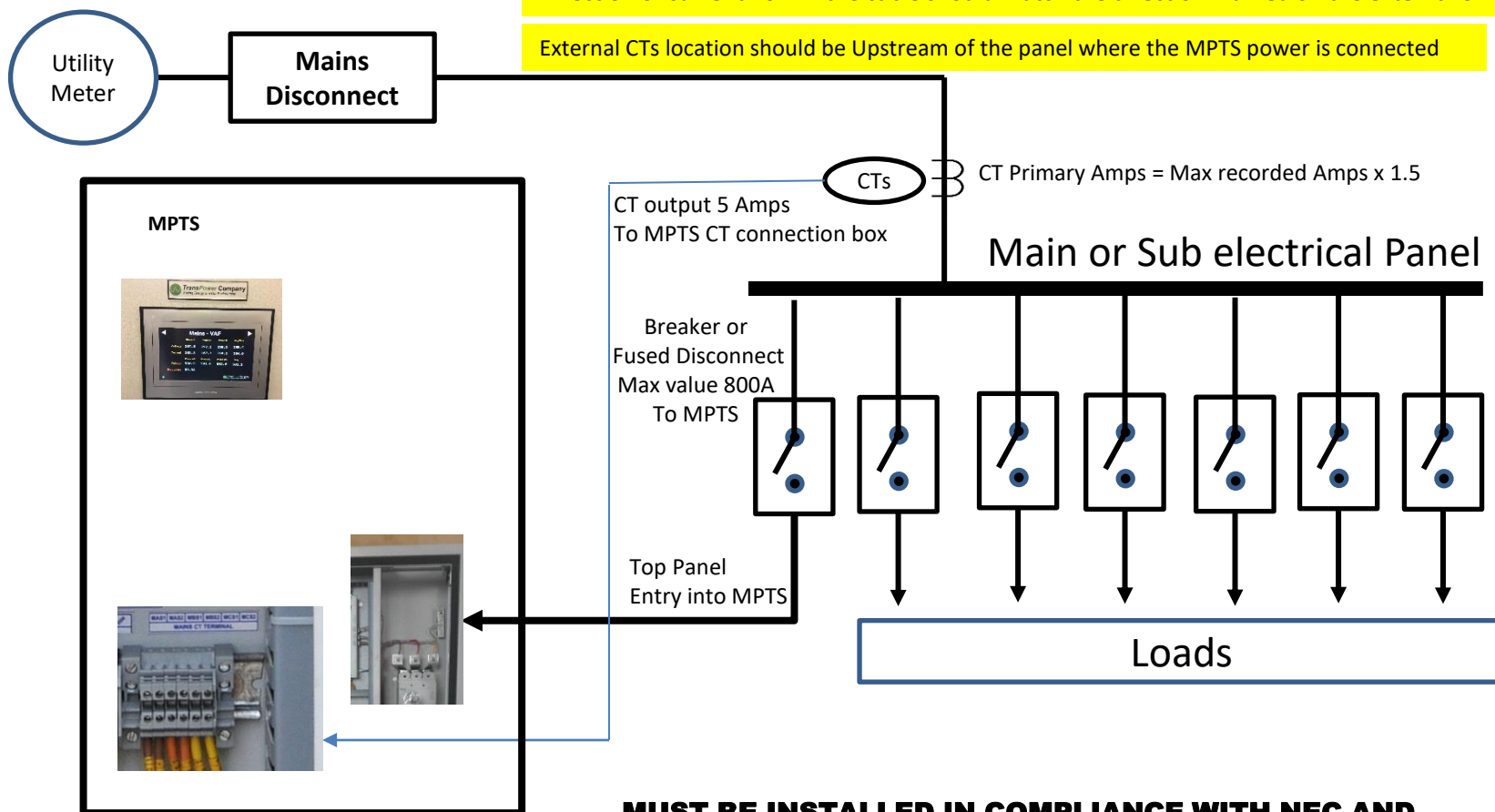
MPTS installed in August 2017

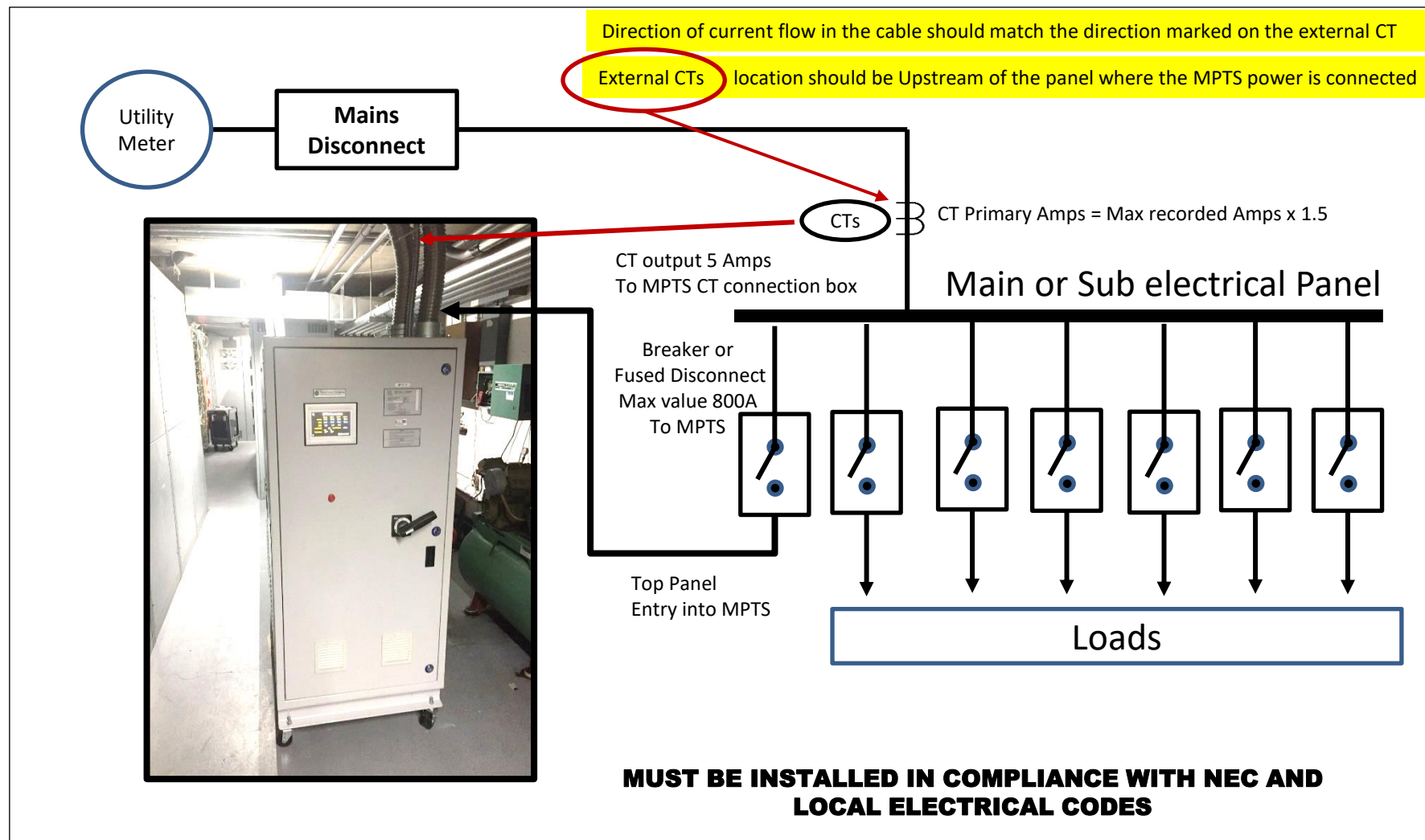
MUST BE INSTALLED IN COMPLIANCE WITH NEC AND LOCAL ELECTRICAL CODES'

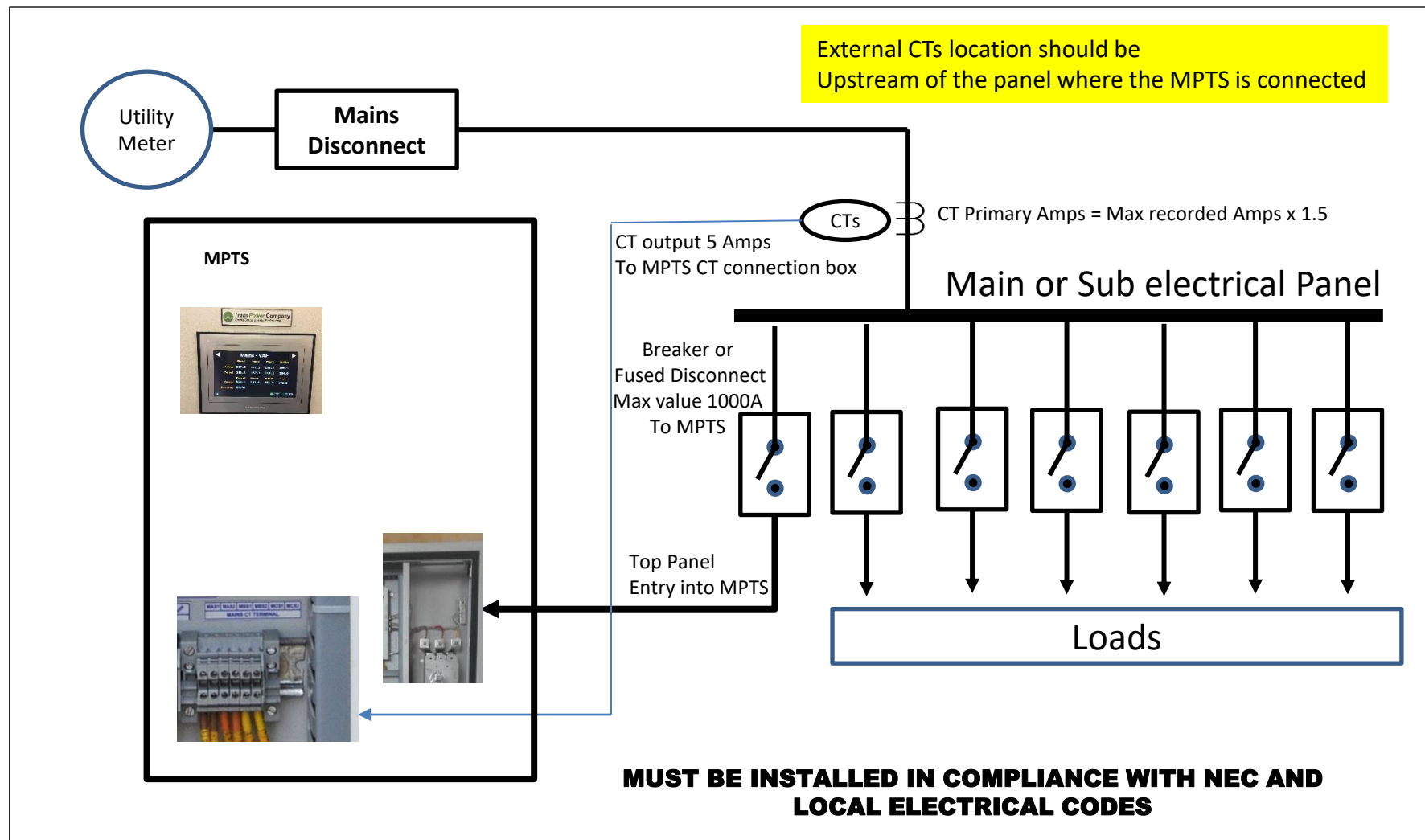
490A/480V/3Ph/400KVA_r

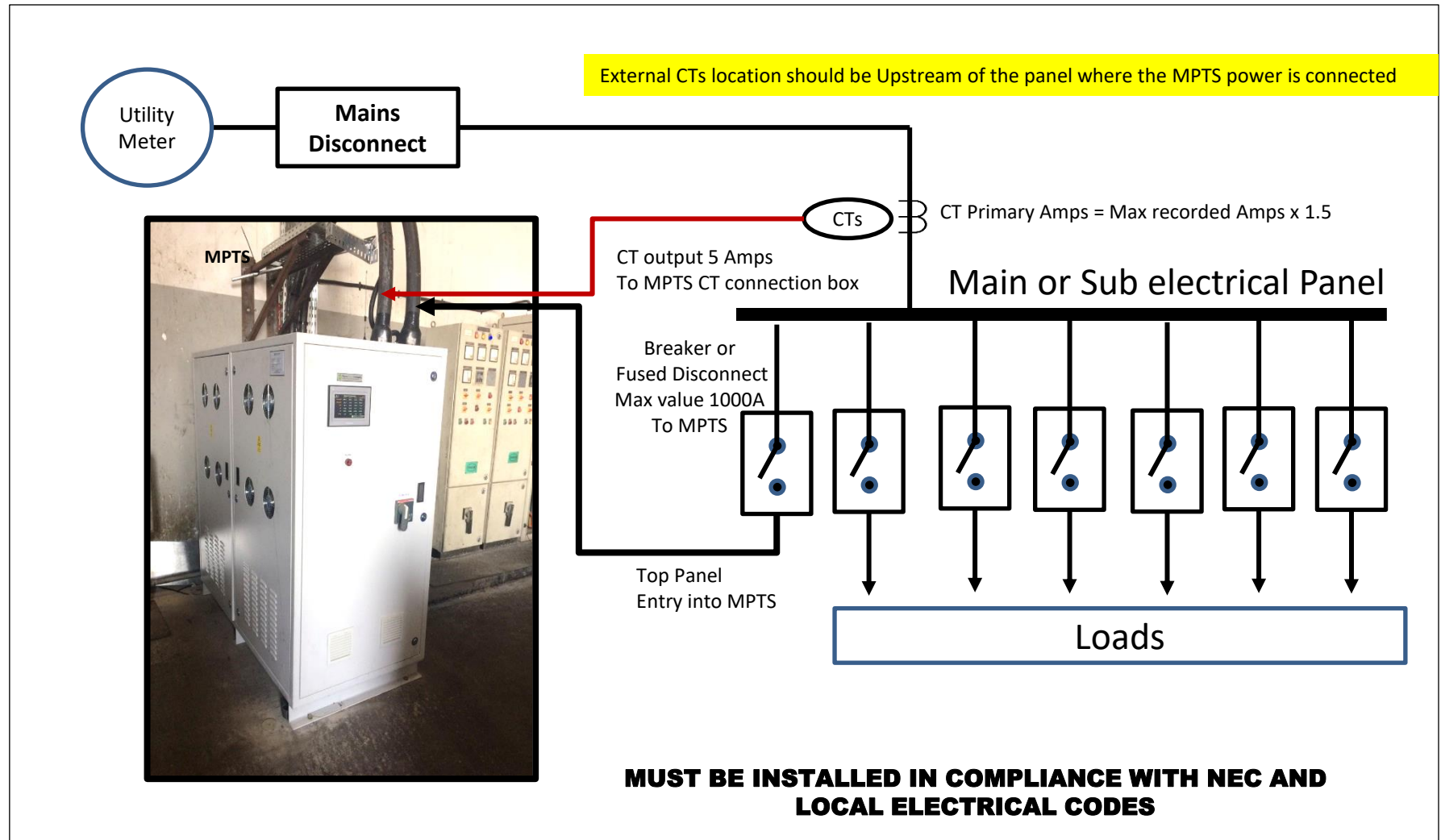
Direction of current flow in the cable should match the direction marked on the external CT

External CTs location should be Upstream of the panel where the MPTS power is connected



490A/480V/3Ph/400KVA_r

720A/480V/3Ph/600KVA_r

720A/480V/3Ph/600KVA_r

Off-line Connection Verification

Conduct a continuity test of each and every connection using a multi-meter to make sure that all the sequences are correct.

If any of the sequences is found to be incorrect, MPTS will not work properly. It is mandatory to confirm all connections.

Pre-commissioning checks

MPTS Devices are shipped with Internal MCCB (s) in 'OFF' position.

It is essential and critical to connect the incoming and outgoing phases / terminals as per code in a correct manner. when phases are interchanged during connection it can result in a safety and equipment hazard. This can also damage the MPTS Device

MPTS installation consists of connecting the supply input consisting of three incoming phases(a,b,c) + the neutral and grounding terminal. Position the unit as per codes with proper clearances for ventilation and access on each side of the unit.

Confirm the Voltage, Load Current and type of connection (Delta-3 wire or WYE - 4 wire) are correct.

Check and confirm that the incoming mains breaker and related electrical system is compliant to the NEC and local code (s).

MPTS must be installed by a TransPower Company certified electrician or engineer and in applications as per manufacturer recommendations. Warranty is void if MPTS installation is not in compliance with the above and the NEC /Local electrical codes

Commissioning and Start-up Procedure

Conduct a final visual check to ensure that it is safe to start commissioning the MPTS (all connections secured, tools removed, safety checks completed as per safety requirements and NEC).

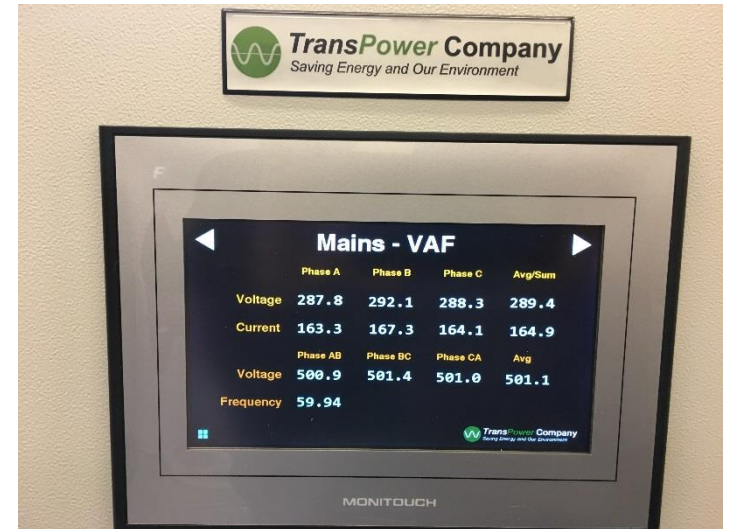
Turn ON all the breakers inside the unit. Close all the access panels on the MPTS and confirm that the panels are all locked and secured shut.

Turn ON the power supply to the panel where MPTS is connected, turn the breaker in the electrical sub panel to ON position to power the MPTS. Turn the connected loads ON

The Multi Function Meter (MFM) controller will indicate the readings of electrical system parameters

Make sure the Voltage, and Current values displayed on the MPTS match the external values on the Fluke Meter

When external CT's are connected the CT ratio information may be required for input into the meter to display correct values



Commissioning and Start-up Procedure

When the MPTS unit is energized, the following will happen:

Multifunction Meter (MFM) will display incoming electrical data.

The ventilation Fans will start. Multi-Function Meter (MFM) checks and displays parameters like Voltage, Current, KVA, KVAR, PF for each phase.

Confirm that the meter is displaying the parameters stated above to make sure that connections are correct.

Record the following parameters :

Date:			Location:		
Time:					
Parameters		MPTS Mains		MPTS–Loads	
Voltage –Volts (L-L)					
Current – Amps (L-L)					
KVA (Total)					
KVA _r (Total)					
Power Factor					
Frequency – (Hz)					

Limited Warranty

TransPower Company warrants to the original purchaser that products sold shall be free from defects in material and workmanship for the warranty period of 12 months / One year from the Date of Purchase.

If buyer claims that a product violates this Limited Warranty, TransPower, upon notice from the buyer, will either examine the product at buyer's site, or issue shipping instructions for return to TransPower at buyer's expense, transportation charges prepaid.

TransPower's sole obligation under this Limited Warranty shall be, at its option, to repair, replace or issue credit for the price of any product which is proved to violate this Limited Warranty.

In no event, whether based on contract, indemnity, warranty, tort (including negligence), strict liability or otherwise, shall TransPower Company be liable to the buyer for special, indirect, incidental or consequential damages whatsoever including, without limitation, loss of profit or revenue.

THIS LIMITED WARRANTY IS THE BUYER'S EXCLUSIVE REMEDY AND TransPower Company HEREBY EXPRESSLY DISCLAIMS ALL OTHER WARRANTIES, EXPRESS OR IMPLIED, INCLUDING THE IMPLIED WARRANTY OF MERCHANTABILITY AND THE IMPLIED WARRANTY OF FITNESS FOR A PARTICULAR PURPOSE.

The foregoing shall constitute the sole remedy of the buyer and the sole liability of TransPower company. This Limited Warranty shall not apply to any product or component thereof which has been repaired or altered by anyone other than Transpower authorized personnel in any manner so as, in TransPower's sole judgment, to affect its serviceability, or to any product that has been subject to alteration, accident, misuse, abuse, neglect or normal wear.

This Limited Warranty shall not apply to products which have been assembled or installed or used in a manner contrary to TransPower's printed instructions, or due to failure to follow Transpower printed instructions for operation and maintenance or installed by a non certified installer.

Any technical assistance provided by TransPower's personnel or representatives in system design or installation is construed to be a proposal and not a recommendation. The responsibility for determining feasibility for use of TransPower's products rests with the user/buyer.

Only the terms expressed in this Limited Warranty shall apply and no agent, reseller, distributor, corporation or individual is authorized to amend, modify or extend this Limited Warranty in any way on resale.

Operation and Maintenance

When MPTS units are turned ON, the reduction in current will indicate that the device is working. The total electrical power quality will improve. This is indicated by the reduction in total electrical power required for the same connected load. At the input or the line side of the MPTS there will be a reduction in current, KVA, KVAR, and improvement in power factor (PF). The meter on the unit will display the performance values. Once the unit is commissioned, turned ON, it will continue to function and perform effectively managing the energy to protect the connected load against surges, spikes, and harmonics. The service, repair, replacement and maintenance costs on the connected load and on other equipment in the electrical network will be significantly reduced.

The MPTS is a self adjusting unit designed for fully automatic operations.

It is recommended to periodically check (three to four times a year or subject to site conditions and during maintenance shut downs) the following after ensuring that the unit is safely isolated or turned OFF.

NOTE: Always de-energize the MPTS unit before maintenance.

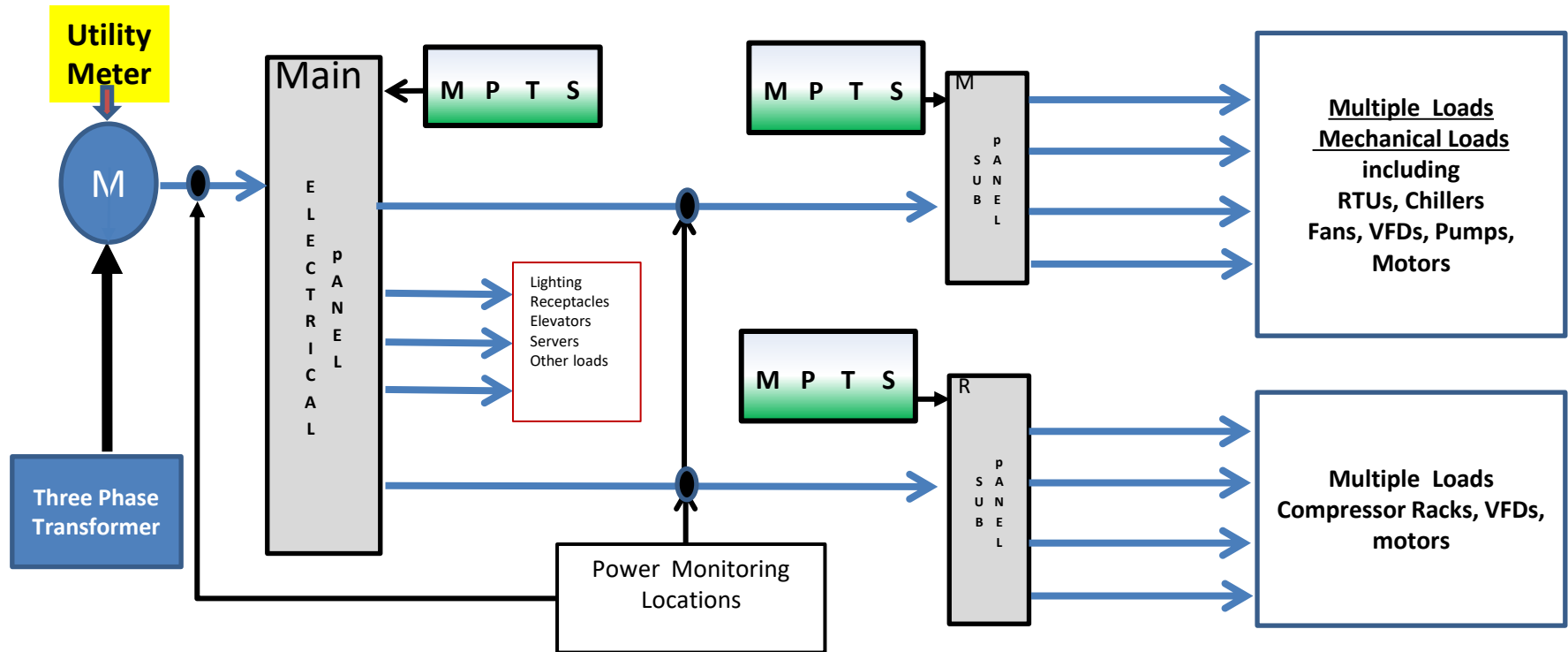
- Depending on the environment and site conditions, the unit should be periodically inspected internally by a qualified electrical technician.
 - It is recommended to open the side panels, visually inspect, and check internal conditions
-

Operation and Maintenance ...continued

- Inspect the vent filters and clean or replace them as required. The filter housing is very easily disassembled and the filters accessed. In case there is accumulation of fine dust and other materials, use a slow vacuum pump to clean the filters.
 - Inspect the surge arrestor periodically; GREEN flag indicates it is normal and RED flag indicates that there was a major surge event and the arrestor failed. The surge arrestor is modular and only the failed module should be replaced.
 - For units that are frequently turned ON and OFF due to intermittent loading or limited time operation, frequent expansion and contraction may loosen contacts. It is important to check these periodically and tighten them (if required) to avoid any failure due to loose terminal or other contacts. Do not overtighten any connections or termination contacts.
 - In case the unit is operated in a dusty environment and if dust/grit collection is noticed during inspection, it is recommended to clean more frequently (site specific timetable) with an appropriate low vacuum suction to clean the unit internally. Compressed air can also be used to keep the inside of the MPTS unit clean.
 - After completing the periodic maintenance procedures, ensure that no tools are left inside the enclosure. Close all sections of the enclosure securely.
-

Typical Layout – One Line Electrical Diagram with MPTS (One or more MPTS device strategically located)

MPTS will improve Power Quality, Solve Power Issues and make building's average Power Factor greater than 0.95



Average Power Factor Existing Values at the Mains (MDP), **without MPTS**, close to the Utility meter, are anywhere from **0.70 to 0.95**
With MPTS the PF will be greater than 0.95. MPTS will maintain the power factor above 0.95 even with changing loads

Power Factors at Sub Panels without MPTS

HVAC	= 0.65 to 0.85
Refrigeration	= below 0.60
Factory equipment	= below 0.60
Elevator and escalators	= below 0.50
Motors	= Below 0.85
Receptacle Loads	= below 0.70
Compressor motors	= below 0.60
Lighting loads	= 0.60 to 0.95

Reduce Watts per Square Foot

Commercial buildings

- Office and Retail
- Banks
- Schools and colleges
- City and County offices

Factory buildings

- Manufacturing and
- warehousing

M P T S - Technology

Science behind the technology:

Maximum Power Transfer Theorem (*Jacobi's Law - 1840*)
focuses on increasing total electrical efficiency by decreasing losses using impedance matching.

MPTS
MAXIMUM POWER TRANSFER SOLUTION
IS THE NEXT GENERATION
POWER QUALITY SOLUTION

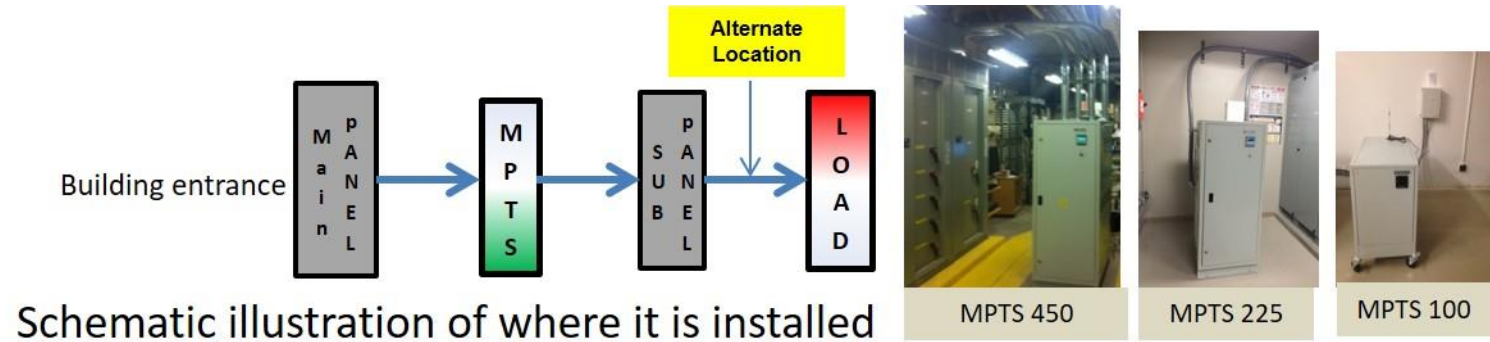
Clean energy technologies
Carbon emissions reduction
Decrease operating costs
Energy efficiency and conservation

MPTS Defined:

MPTS, maximum power transfer solutions, are next generation power quality products that **decrease the Electrical Demand, Total Electrical Power requirement or Generation(KVA) and the carbon foot print without replacing/retrofitting inefficient electrical and mechanical equipment**



For additional information, CONTACT your supplier



MPTS H240



MPTS H240



MPTS H490



MPTS H720